

OceanEmbed — Research Synopsis

Satellite Embedding-Based Deep Learning Framework for Subsurface Ocean Temperature Reconstruction | Smart India Hackathon — Problem Statement Research Brief

This is a working survey of the existing literature on reconstructing subsurface ocean temperature from surface satellite data, organized around the OceanEmbed PS requirements: North Indian Ocean (5°N–30°N, 45°E–105°E), 0.25° daily fields, 15 standard depth levels (0–1000 m), and a framework built from CNN / ViT / Autoencoder / GNN / attention architectures. The intent is to map who has already worked on this problem, what they used, and which of their tools or ideas are worth pulling into our own pipeline — not to propose our own architecture.

1. Foundational and global-scale reconstruction models

- **Su et al. 2022 (DORS dataset)** — ConvLSTM combined with ARGO-derived labels; produced the first large-scale global subsurface temperature dataset (0–2000 m, 23 depth layers, 1° resolution). Validated against EN4, IAP, Ishii and ARMOR3D — a good template for how to evaluate a reconstruction against multiple independent products.
- **Zhang et al. 2024 (Western Pacific, improved ConvLSTM)** — standardized to 0.25°, same as our PS. Key trick: iterative depth prediction, where the 50 m prediction feeds into models for 100–500 m layers. Reported ~11% RMSE reduction and ~9% R^2 increase over standard ConvLSTM — directly relevant to our 15-depth-level output.

2. Transformer and attention-based models

- **Convformer (2024)** — combines ConvLSTM with a Vision Transformer plus spatial and temporal attention, built specifically for multi-source satellite → subsurface temperature/salinity. Reports ~0.35°C RMSE in the tropical Pacific, though accuracy drops near the thermocline.
- **TemproNet** — transformer-based seawater temperature model; cites **He et al. 2021**, a variational reconstruction set specifically in the Northern Indian Ocean (SST/SSS/SSH inputs, captured mesoscale eddies well in the Arabian Sea and Bay of Bengal). This is the one prior study actually located in our region.
- **Dual-attention CNN (Bay of Bengal)** — used for mixed layer depth rather than temperature profile, but regionally relevant for capturing barrier-layer effects specific to the Bay of Bengal freshwater cap.

3. CNN-based architectures

- **Dual-Path CNN (DP-CNN, 2023)** — two parallel processing paths to avoid losing fine spatial detail that a single-path CNN would smooth over. Tested in the South China Sea (ST $R^2 \approx 0.93$, RMSE $\approx 0.31^\circ\text{C}$).

- **Physics-constrained CNN inversion (South China Sea)** — adds a heat-budget balance term to the loss function so the model respects a basic physical constraint rather than treating reconstruction as pure regression.

4. Volumetric and multi-scale reconstruction

- **Attention-enhanced 3D U-Net++ (2026)** — the most operationally relevant paper found. Explicitly addresses the fact that SST and SSH arrive in near real time while surface wind lags 1–3 days and SSS lags 3–7 days. Uses transfer learning: pretrain on monthly ARGO-gridded data, then fine-tune on daily SST/SSH plus GLORYS2V4. We will hit this exact latency mismatch with our own input variables.
- **Adaptive spatiotemporal clustering framework (2026)** — a model-agnostic training trick: restructuring how training samples are clustered spatiotemporally before training improved DP-CNN, Attention U-Net, and ViT backbones by 12–27% RMSE. Cheap to borrow regardless of which architecture we choose.

5. Physics-guided and physics-informed learning

- **Physics-guided ConvLSTM (2023)** — incorporates ocean stratification and physical constraints into training rather than treating the model as a black box; reports improved accuracy and physical consistency.
- **OG-PINN (Observation-Guided Physics-Informed Neural Network)** — reduces RMSE by up to 5% and improves correlation by up to 7% over standard PINNs, specifically by dynamically reducing physical-loss weight in dynamically complex regions such as upwelling zones — relevant to the Arabian Sea's seasonal upwelling.

6. Graph neural network approaches

- **STGAT (Spatiotemporal Graph Attention Network)** — reconstructs 21 depth levels (20–1941 m) from SLA, SST, SSS, and surface wind, validated against GLORYS12V1 and EN4. Closest GNN match to the PS's own GNN mention.
- **MG-GCNN (Multi-Granularity Graph Neural Network)** — treats observation points as heterogeneous graph nodes and incorporates bathymetry as a geometric prior — relevant given the NIO's strong bathymetric control (continental shelf, Andaman arc).
- **3D-DSTGN** — builds static and dynamic graph structures for full 3D temperature fields down to ~2000 m, combining graph convolution with multi-scale temporal convolution.
- **SeaCast** — a GNN originally built for the Mediterranean, later adapted to the North Atlantic — a useful reference for how a regional GNN gets transplanted to a new basin.
- **CLOINet** — blends classical Optimal Interpolation with a learned latent "cluster" space, bridging traditional objective analysis (the method INCOIS's own gridded ARGO product uses) with embedding-based deep learning.

7. Embeddings and autoencoders — closest match to the PS's own framing

The PS explicitly asks for “compact satellite embeddings,” which makes this the most directly relevant thread despite being a smaller body of work.

- **TS-Cast (2026)** — compresses surface variables into a 512-dimensional latent vector, decodes through a U-Net-style architecture into temperature and salinity profiles, with a second output head predicting depth-dependent uncertainty. Close to a working template for “surface embedding → subsurface profile.”
- **Ulmo** — a probabilistic autoencoder on MODIS SST patches; compresses 128×128 cutouts into a 512-dim latent vector and fits a normalizing flow on top to flag anomalous ocean states. Reusable idea: flagging regions where reconstruction confidence is low.
- **MAESSTRO (2024)** — masked autoencoder plus ViT for SST reconstruction under cloud occlusion; RMSE under 0.2°C even with 80% of the image masked. Directly useful for the gap-filling/harmonization step before any embedding stage.
- **PDAE (Probabilistic Deep Autoencoder)** — forecasts subsurface temperature up to two months ahead using an autoencoder plus Bayesian optimization.
- **Conditional GAN (Zhang et al. 2023)** — generates subsurface temperature conditioned on SST/SLA instead of random noise; trained on HYCOM reanalysis, fine-tuned on satellite plus CORA 2.0 data.
- **AlphaEarth Foundations (Google DeepMind, 2025)** — not ocean-specific, but the most visible real-world example of the exact “surface embedding → subsurface property” pattern the PS describes. Compresses multi-source Earth observation data into 64-dimensional per-pixel embeddings; a companion study applies these to subsurface geological estimation (seismic velocity, ground temperature) in the US. Worth knowing conceptually; not directly reusable since it isn't trained on ARGO/GLORYS or ocean variables.

8. Other techniques worth noting

- **DINEOF** — an older, open-source (GitHub) gap-filling method based on empirical orthogonal functions; useful as a lightweight, fast baseline before investing in deep learning.
- **Sub-surface velocity reconstruction via Self-Organizing Maps** — open MATLAB code on GitHub; a non-deep-learning alternative worth knowing about for comparison.
- **SHAP** — feature attribution, not ocean-specific; several papers use it to explain which surface variable (SST vs. SSH vs. wind vs. SSS) drives a given depth's prediction — doubles as a sanity check.

9. Data and validation sources across the literature

Source	What it provides	Note
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ARGO / INCOIS Gridded ARGO in-situ T/S profiles	INCOIS product uses DIVA method, gridded at 0.25° for the tropical
GLORYS (Copernicus Marine) 3D ocean reanalysis	Named directly as the training target in the PS
EN4, IAP, Ishii, ARMOR3D Independent validation products	Used across nearly every paper to cross-check reconstructions aga
CMEMS / Copernicus Marine Reanalysis + satellite/in-situ products	Accessed via the free Copernicus Marine Toolbox — no download o
OISST, CCMP SST and surface wind products	Common satellite input choices
INCOIS Live Access Server Regional NIO products, MLD, D20, heat budget	Indicate specific data service named in the PS

10. Practical tooling stack

- **xarray + Dask + Zarr** — standard stack for multidimensional NetCDF-style ocean data (time × depth × lat × lon) without loading everything into memory at once.
- **Copernicus Marine Toolbox** — free, no quota, programmatic subsetting of GLORYS and other CMEMS products by variable/region/date/depth (Python API or CLI).
- **PyTorch** — framework used across nearly all cited architectures (CNN, ConvLSTM, ViT, U-Net, GNN via PyTorch Geometric).
- **OceanBench** — an actual benchmarking framework for ocean ML, useful for a standardized evaluation methodology instead of inventing metrics from scratch.

Reading priority

If narrowing down to a handful of papers to actually work through in depth:

- **TS-Cast** — closest existing match to “surface embedding → subsurface profile with uncertainty.”
- **STGAT** — the graph-based alternative to the same idea.
- **Attention 3D-U-Net++** — handles the real-time/lagged-input problem we will hit with SSS and wind data.
- **He et al. 2021** — the one paper actually set in our region.
- **Adaptive spatiotemporal clustering framework** — a cheap technique reported to boost several architectures at once.
- **MAESSTRO** — for the preprocessing/gap-filling step before anything else runs.

Everything else above is worth keeping as reference material for the writeup — it shows the judges the field was actually surveyed rather than one architecture picked at random.